

Diwita Banerjee

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 5+ years of experience building and deploying scalable Machine Learning, LLM, and Generative AI solutions across enterprise applications. Expertise in Retrieval-Augmented Generation (RAG), NLP, transformer models, predictive analytics, and MLOps using Python, LangChain, Hugging Face, AWS, and GCP. Proven ability to optimize model performance, reduce inference latency, and deliver AI-driven business impact through production-grade solutions.

EDUCATION

Master of Science in Computer Science (GPA: 3.73/4.00)

Aug 2023 – May 2025

George Mason University, Fairfax, VA

TECHNICAL SKILLS

Languages & Database Tools: Python, Java, MATLAB, SQL, MySQL, SSMS, PostgreSQL, Redis

Libraries & Frameworks: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, NLTK, Hugging Face Transformers, LangChain

Machine Learning: Linear & Logistic Regression, Decision Trees, Random Forest, Naive Bayes, A/B Testing, Feature Engineering, Model Evaluation

Deep Learning & NLP/LLMs: CNN, RNN, LSTM, NLP, Large Language Models (LLMs), BERT, GPT-3, Prompt Engineering, Retrieval-Augmented Generation (RAG), FAISS, RoBERTa, PEFT / LoRA, Prompt-Injection Defense, Model Robustness Evaluation, Bias and Fairness Evaluation

Cloud: AWS (EC2, SQS, SNS, Lambda, CodeDeploy, CloudWatch, API Gateway), GCP (Vertex AI, Google Cloud Storage)

Statistical Techniques: Hypothesis Testing, Data Modeling, Data Visualization (Tableau, Power BI), Experiment Design

MLOps & Deployment: FastAPI, Airflow, MLflow, PySpark, Docker, Jenkins, Kafka, OAuth, CI/CD, Drift detection, Model monitoring / observability

Certifications: GCP (Professional Data Engineer, Associate Cloud Engineer, Cloud Digital Leader), Azure (AI Engineer Associate, AI Fundamentals, Data Fundamentals), AWS (AI Practitioner, Machine Learning Specialty)

PROFESSIONAL EXPERIENCE

DoorDash | Machine Learning Engineer

Dec 2024 – Present

- Designed and deployed LLM-based RAG systems using LangChain, FAISS, Hugging Face Transformers, and fine-tuned transformer models such as BERT, RoBERTa, and Llama using PEFT/LoRA for merchant support, customer sentiment analysis, and delivery operations, reducing query resolution time by 35% and manual effort by 30%.
- Built and optimized machine learning models for delivery time prediction, order demand forecasting, and customer personalization, improving ETA accuracy and increasing operational efficiency across logistics workflows.
- Developed production-grade FastAPI microservices and automated end-to-end ML pipelines for training, validation, deployment, and monitoring using AWS, Airflow, MLflow, Docker, and Jenkins, reducing inference latency by 40% and model release cycles by 50%.
- Implemented real-time monitoring and drift detection for recommendation and prediction models using PySpark, CloudWatch, and model observability tools, ensuring consistent production performance and reliability.
- Collaborated with data scientists, product managers, and software engineers to add personalization and automation features to customer and merchant platforms using Hugging Face Transformers, Keras, and custom Python feature-engineering pipelines, which increased user engagement and accelerated rollout of personalized features

George Mason University | AI Research Assistant

May 2024 – Nov 2024

- Developed fairness and bias evaluation pipelines for multimodal AI systems (Stable Diffusion, LLaVA, InstructBLIP), measuring performance across demographic groups and identifying critical bias patterns.
- Built a medical visual question. Answering (VQA) system using MIMIC-IV EHR data to support clinician queries, reducing query resolution time by 25% in controlled user evaluations.
- Optimized large-scale training and inference pipelines on HPC clusters by tuning batch sizes, parallelism, and memory usage, reducing end-to-end runtime by 40%.
- Designed and tested prompt-injection defense and robustness pipelines to improve the safety and reliability of generative AI systems.
- Published research findings and collaborated with faculty and students to translate experimental AI systems into practical, deployable prototypes.

Cognizant Technology Solutions | Machine Learning Developer

Nov 2021 – Jul 2023

- Built predictive machine learning models for lead, case, and renewal scoring to help sales and service teams prioritize work, improving business decision quality and operational efficiency.
- Automated nightly training, validation, and batch inference pipelines to ensure models stayed up to date with fresh data at enterprise scale.
- Developed dashboards to track model precision, recall, adoption, and drift, with automated alerts when performance degraded.
- Integrated ML services into production systems via secure REST APIs built with FastAPI, event-driven pipelines using Kafka, and OAuth authentication, which reduced integration time and improved deployment reliability.
- Implemented CI/CD pipelines with automated testing and rollback strategies to ensure stable and repeatable ML deployments.

LTIMindtree | Junior AI Developer

Jun 2020 – Oct 2021

- Designed and built predictive analytics services using Python and enterprise backends, enabling data-driven decision-making across multiple business units.
- Exposed ML models through REST APIs for seamless integration with enterprise applications.
- Optimized backend services to reduce response latency by 39% through efficient model serving and system design.
- Automated defect root-cause analysis workflows using ML techniques, reducing issue resolution time by 30%.
- Built data-driven UI components to visualize AI insights, improving user adoption and operational visibility.